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Bridging the short-term and long-term dynamics of economic structural change

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Economic development hinges on structural change, that is, transformations in what an economy produces. The field of economic complexity has investigated this process through two related but distinct branches: one studying how economies diversify, the other how the complexity of an economy is reflected in its output. However, a formal connection between these approaches, and their relationship to classic accounts of structural transformation (for example, from agriculture to manufacturing), remains unclear. Here we introduce a simple dynamical model that links these perspectives through one core idea: economies diversify preferentially into activities related to those they already do. Studying this model yields three main results: It generates quantities resembling economic complexity metrics, suggests these metrics summarize long-term structural change rather than directly infer an economy's complexity, and reproduces stylized facts of development. Our framework formally connects the field's conceptual strands, bridges short and long timescales of change, and adds granularity to classic descriptions of development.

An economy's prosperity is tied to the economic activities it can develop[1]. Whereas places like the city of London and the country of Japan have grown economies containing diverse and high-value industries, others struggle to shift out of a narrow range of sectors with low economic returns. Seeking to understand why, the field of economic complexity has developed two branches of investigation. The first asks how economies enter new economic activities and finds that such economic diversification is biased toward new activities that are related to ones already in place. This tendency has been corroborated in work on industry clusters[2,3] and related diversification by individuals[4,5], firms[6–8], regions[9–13], and countries[14–16]. Recently, these tendencies have been referred to collectively as the *Principle of Relatedness*[17].

A second branch has explored a conceptualization of development in which countries grow by gaining *capabilities*. Different combinations of capabilities enable the production of different products, with more complex economies possessing more capabilities and more complex products requiring more. Capabilities are sometimes seen as hidden variables, information about which must be inferred from country output baskets, and thus a key objective has been to develop *complexity metrics* that extract complexity from data[18–30]. These metrics aim to estimate the complexity of places and products from the structure of bipartite networks that link places to those products they make in economically-significant quantities (in other words, suggesting the location has the needed capabilities to engage competitively in a product market). The earliest complexity metrics were the Economic Complexity Index (ECI)[18] and country Fitness[19], and others (such as GENEPY, the generalized Economic comPlexitY index) have been proposed in their wake[23,24,26,27,29,30].

While both branches have found success, a lack of theoretical underpinnings risks limiting the development of this research agenda and its interpretability for policy work. Currently, the two branches of work are linked mainly by the narrative of capabilities rather than by precise mathematical connections. In addition, both branches

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constitute an approach to structural change, a research topic stretching back decades, yet it is unclear how this work relates to classic findings such as the Kuznets facts[31]. Here we present a simple modeling framework that links these topics, connecting the two branches of economic complexity to one another and to stylized facts of development. Our results relate economic complexity to classic work, matching established observations while adding detail and nuance to them. In addition, our results suggest prominent complexity metrics may mainly describe overt changes that economies undergo during development, rather than infer complexity per se.

We start by making explicit the dynamics that are implicitly used by models in the first research branch to describe economic diversification according to the Principle of Relatedness (PoR). PoR models describe economic structural change on short (for instance, year-to-year) time scales. We analyze how a country that obeys the PoR would evolve over the long run. While the PoR is an incomplete description of economic transformation, we can nevertheless derive the long-term predictions that are implicit in how the PoR has been formulated. We show that the PoR implies the importance of at least two kinds of long-run changes to an economy's basket of activities. One entails a change in the diversity of activities (for example, the number of product categories in which an economy exports). The other involves changes in their relative mix or composition. We derive a pair of coordinates that track these changes, one associated with diversity, the other with a particular pattern of shifts in an economy's activity basket.

We show how these coordinates re-express a traditional understanding of structural change, resonating well with how economists have long described economic development: Countries diversify as they rise through lower and middle stages of income[32,33], and undergo compositional changes that include shifting out of labor-intensive forms of agriculture and into industrial sectors[31,34]. To demonstrate this agreement, we apply the coordinates that emerge from the PoR to data on global trade patterns, using them to describe 56 years of change in the export baskets of about 250 countries and regions. We find that the dominant movements of these coordinates are intuitive and consistent with a well-documented pattern of development—countries diversify into more products while simultaneously shifting out of agricultural products toward manufactured goods (see, for instance[32,35]). (Though our approach makes such statements more precise, describing this shift in granular terms of the specific industries abandoned or developed along long-run growth paths.) The results support the empirical relevance of these coordinates and an interpretation of them as summary measures of economic shifts associated with structural change. They also demonstrate that transient, year-to-year dynamics expressed by the PoR contain substantial information about the long-term, permanent development of economies.

We next observe a correspondence between the coordinates implied by the PoR and economic complexity metrics. Empirically, nearly all complexity metrics fall in one of two groups (Fig. 1). This in itself is somewhat surprising, as different complexity metrics are motivated from a wide range of theoretical arguments[18–20,23–30]. Examining these groups, one contains metrics that strongly correlate with an economy's diversity, and includes the country Fitness metric, while the other group captures compositional information about an economy, and includes the ECI. These two groups of metrics correspond numerically and theoretically to the two coordinates implied by the PoR. That is, complexity metrics empirically come in two main types, and these types track the two kinds of long-run evolution that are expected of an economy whose short-run changes are described by the PoR. Given this, we propose a connection between the two main branches of work in economic complexity: Complexity metrics describe the overt, long-term changes to an economy's basket of activities that are implied by the Principle of Relatedness.

We thus find that a single dynamical model implies the importance of coordinates resembling both kinds of complexity metrics.

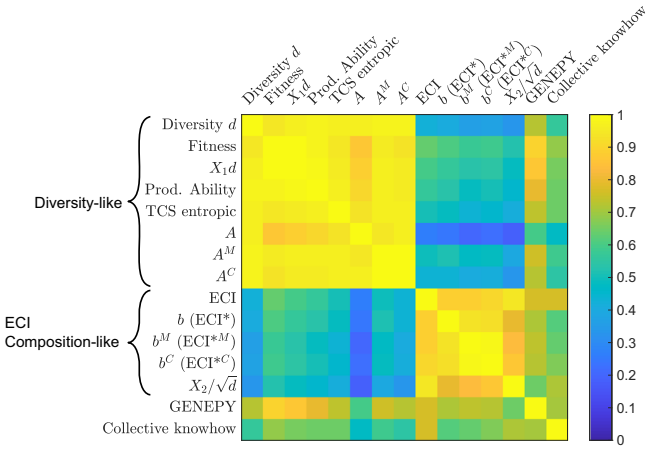

Fig. 1 | Spearman rank correlations. Rank correlations among complexity metrics and the coordinates that emerge from our dynamical modeling framework. Diversity here is the count of activities in which a place has a revealed comparative advantage above 1, a common measure of the number of activities that a place performs competitively[18,19], and often taken to have a close association with an economy's underlying complexity. Proposed complexity metrics include the Economic Complexity Index (ECI)[18], country Fitness[19], Production Ability[26], GENEPY[27], Collective knowhow[30], and the entropic measure of Teza, Caraglio, and Stella[24] (here labeled TCS entropic). X_1 and X_2 are the first and second components of the GENEPY metric[27]. A, A^M, and A^C are different variations of the first coordinate that emerges from our dynamical framework, based on different literature approaches for estimating proximity between activities b, b^M, and b^C are the corresponding variations of the second coordinate that emerges from our framework. All complexity metrics and other quantities were computed for the year 2016 using UN Comtrade data[47].

Crucially, it does this without using the conceptual narrative of country capabilities that motivated these metrics. Two corollaries of this point are noteworthy. First, the most prominent complexity metrics, the ECI and country Fitness, are usually viewed as competing methods to infer information about an economy's capabilities. Yet our results suggest they are not in competition, but measure distinct and complementary properties of a country's activity basket, each being incomplete without the other. Second, neither necessarily infers hidden information about capabilities, but may just summarize overt information about the diversity or composition of an economy's output. Complexity metrics have become widely used for economic policy[36], and our results raise the question of how policy frameworks would need to be adjusted if these metrics do not infer complexity but instead summarize long-run changes in an economy's portfolio of activities.

Our approach to modeling structural change exploits standard dynamical systems methods and leads to natural inter-connections between the PoR, complexity metrics and classic observations. This cohesion makes the framework an appealing starting point for further studies. Our emphasis on structural change may also shed light on the use of complexity metrics to study regional and city development[9–13], where the link between complexity measures and the sophistication of industries has sometimes proved tenuous. Our work may help explain these difficulties and suggest solutions that shift interpretation from complexity to economic transformation.

Results

The principle of relatedness as a dynamical model

A well-established finding is that economic activities colocate in predictable ways[9,15,37]. Such colocation patterns are often intuitive. For example, a city, region, or country that makes cloth garments is also likely to make other textiles (such as knitted garments). These geographical co-occurrences are assumed to indicate that two activities are related (meaning that they depend on similar underlying

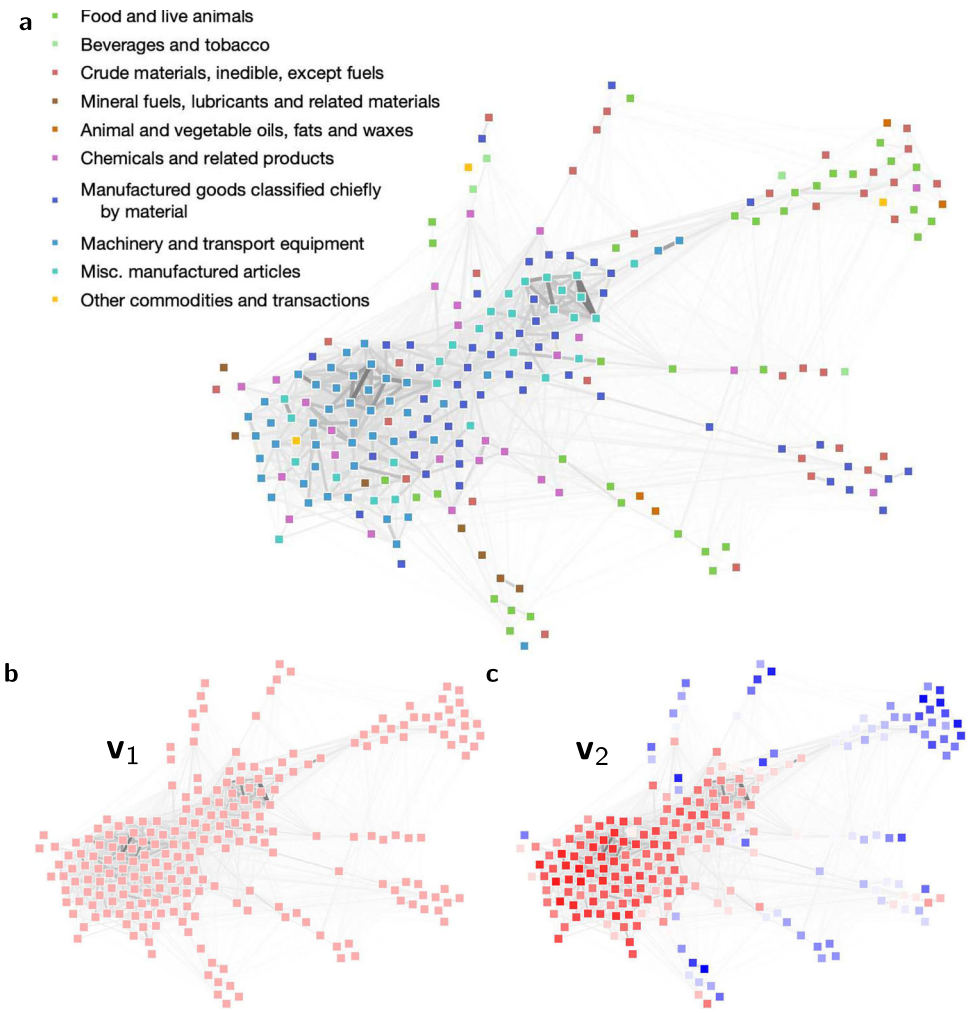

Fig. 2 | Network visualizations of relatedeness between tradable products. a To construct the network above we use an approach similar to Hidalgo et al. (2007)[15]. Using UN Comtrade data[47] we compute the revealed compared advantages (RCAs) of each country in each product p, and obtain the conditional probability across countries $Q_{pp'}$ that a country has an RCA greater than 1 in good p given that it also meets this condition for p'. We then compute relatedness between products p and p' as $\Phi^M_{pp'} \equiv \min(Q_{pp'}, Q_{p'p})$. For other approaches to constructing the network, see, for instance, Ref. 16,64. **b, c** The first two right eigenvectors of the row-normalized proximity matrix $\tilde{\Phi}_{pp'} = \Phi_{pp'}/\sum_{p''}\Phi_{pp''}$. The first eigenvector is simply a uniform vector. For the second eigenvector, red nodes are positive entries while blue nodes are negative entries.

capabilities), and based on this assumption, various measures have been developed to infer the *relatedness* of different activities. For concreteness, we focus on relatedness between exported products as inferred from co-occurrences within country export baskets. Exports are often studied because harmonized export data is available across countries of very different levels of development, and because exporting competitively signals the existence of well-developed production abilities. Studies typically measure activity in an export using the Balassa index of revealed comparative advantage (RCA)[38], $R_{cp} = \theta_{cp}/\theta_p$, where θ_{cp} is the share of country c's exports devoted to product p, and θ_p is the share of p in world exports. RCAs are often treated as measures of inferred ability in exported products[39,40], and these quantities have also been used to characterize ability in activities besides exports (where the meanings of θ_{cp} and θ_p are adjusted appropriately). Here we follow suit, although it is not an intrinsic requirement of our model to measure ability by RCAs. (See "Methods" for further discussion of RCAs).

The relatedness—also called proximity—of products p and p' is then computed by a measure of similarity between the p and p' columns of the matrix $R = [R_{cp}]$. That is, p and p' are taken to be related if high ability in p co-occurs with high ability in p' across locations. As an example, Hidalgo et al. (2007)[15] say a country has significant ability in a product if its RCA in the product exceeds 1. The authors then compute the conditional probability $Q_{pp'}$ that country c has $RCA > 1$ in good p, given that it also has $RCA > 1$ for p'. The proximity $\Phi_{pp'}$ between products p and p' is taken to be the lesser of $Q_{pp'}$ and $Q_{p'p}$: $\Phi^M_{pp'} \equiv \min(Q_{pp'}, Q_{p'p})$. Using this or other measures, proximities are often visualized as a network (as in Fig. 2a).

Crucially, proximities predict changes in an economy's activities. Empirical studies show that development of high ability in an exported good tends to be preceded by high ability in related goods[15]. A typical regression setup[11,13,41,42] to explore this effect models future ability in a product p as a function of the *density* around it, defined as the average proximity-weighted ability that an economy has in other products:

$$R_p(t+1) = b_1 R_p(t) + b_2 \sum_{p'} \frac{\Phi_{pp'}}{\sum_{p''}\Phi_{pp''}} R_{p'}(t) + \varepsilon_p(t). \quad (1)$$

An alternate setup[16] contains two steps, with ability in product p first regressed on density in the same time period, establishing the existence of systematic correlations in which economic activities co-occur. Residuals from this regression are then used to forecast appearances of comparative advantage in future periods. Combining two such regressions analytically also leads to an expression in the form of

Eq. (1). These analyses overwhelmingly find a robust, positive statistical association between density and future growth and diversification, bolstering the idea that relatedness measures capture underlying similarities between activities that make some transitions easier to achieve than others, influencing the direction in which an economy develops.

In general, the PoR treats diversification as a process where activities spread over such relatedness networks (with the network typically taken to be fixed in some year). An example is products that spread on the network in Fig. 2. Following Eq. (1) we can make the dynamical model that the PoR implies more explicit. The results that follow use network concepts (see for instance[43]) and the dynamical systems method of eigenmode decomposition (see for example[44]). A given economy has various levels of ability in different activities, which are given by elements of a vector $\mathbf{R}(t)$. We call this the economy's activity basket. These abilities evolve as the economy grows or shifts into new parts of the network. Using Eq. (1) as a guide, a simple model of the diversification process is

$$\dot{\mathbf{R}}(t) = \gamma(t)\mathbf{R}(t) - \frac{1}{\tau}L_\Phi \mathbf{R}(t), \tag{2}$$

where $\gamma(t)$ is an arbitrary growth rate, L_Φ is a graph Laplacian, and τ is a time scale. The two terms on the right capture two kinds of changes. An economy's abilities can rise or fall as a whole, and they can shift according to the Laplacian term to weight activities differently, changing which ones receive the most emphasis.

The Laplacian term reflects an implicit choice of empirical specifications of the PoR as instances of consensus dynamics (see, for instance,[44]). The Laplacian L_Φ captures the structure of the network and governs shifts between activities, taking the form $L_\Phi = I - \tilde{\Phi}$ where $\tilde{\Phi}$ is a proximity matrix with rows normalized to sum to 1, $\tilde{\Phi}_{pp'} \equiv \Phi_{pp'}/\sum_{p''}\Phi_{pp''}$. With this assumption, a discrete-time approximation of Eq. (2) in index form reads

$$\frac{R_p(t+\Delta t) - R_p(t)}{\Delta t} \approx \gamma(t)R_p(t) + \frac{1}{\tau}\left(\sum_{p'}\tilde{\Phi}_{pp'}R_{p'}(t) - R_p(t)\right), \tag{3}$$

which is readily compared to Eq. (1). Ability in activity p changes between periods in a way that depends on ability in nearby products. When the average ability in activities around p ($\sum_{p'}\tilde{\Phi}_{pp'}R_{p'}(t)$) exceeds that in p itself ($R_p(t)$), ability in p rises. The last term in Eq. (3) can also be negative, corresponding to a decline in ability. In addition, abilities have the freedom to rise or fall as a whole because of the term $\gamma(t)R_p(t)$.

We can integrate Eq. (2) over time to give

$$\mathbf{R}(t) = A(t)P(t)\mathbf{R}(0), \tag{4}$$

where the dynamics are now captured by the scalar $A(t)$ and the matrix $P(t)$. $A(t) \equiv e^{\int_0^t \gamma(s)ds}$ is a shift factor that captures accumulated growth in overall ability up to time t, scaling $\mathbf{R}$ up or down as a whole. $P(t) = e^{-L_\Phi(t/\tau)}$ is a stochastic matrix that transforms $\mathbf{R}$, changing the relative emphasis on different activities.

Implementing this model involves some practical considerations. First, RCAs are commonly transformed to weaken the influence of extreme values[16,35,45,46]. When we use RCAs as a measure of ability, we will do so as well, but note that this does not change the model in any fundamental way. Second, we need to choose how to operationalize the network of relatedness $\Phi_{pp'}$ between activities. Studies have used a variety of proximity measures. We consider several options, finding similar outcomes, as we discuss later.

Deriving a coordinate system to describe an economy's activity basket

Dynamical models are frequently analyzed in terms of their dynamical modes, called eigenmodes in linear models. Applying eigenmode decomposition to the model above leads to a coordinate system that can be used to describe the evolution of an economy's activity basket. Let $\mathbf{v}_\mu$ be the μth right eigenvector of L_Φ and let κ_μ be its eigenvalue. Ordering eigenvalues from least to greatest, the first right eigenvector has $\kappa_1 = 0$ and is a uniform vector of real positive numbers. It can be taken to be a vector of 1s, $\mathbf{v}_1 = \mathbf{1}$. Higher-order eigenvectors have $\kappa_\mu > 0$, and contain a mix of elements with positive and negative real parts. (Depending on the proximity matrix used, some eigenvectors could contain imaginary parts, but this would pose no difficulties in interpreting modes; see Supplementary Note 1.) The right eigenvectors of L_Φ form a basis for the vector space of activity baskets $\mathbf{R}(t)$ and so we can write this vector as a linear combination $\mathbf{R}(t) = \sum_\mu c_\mu(t)\mathbf{v}_\mu$. Letting $\mathbf{w}_\mu$ be the μth left eigenvector of L_Φ, the coefficients $c_\mu(t)$ can be computed by exploiting the biorthogonality of left and right eigenvectors, $\mathbf{w}_\mu^\dagger \mathbf{v}_\nu = \delta_{\mu\nu}$, giving $c_\mu(t) = \mathbf{w}_\mu^\dagger \mathbf{R}(t)$.

Following the usual steps of eigenmode decomposition, we decompose $\mathbf{R}(t)$ to separate different modes of change by their time scales. Plugging $\mathbf{R}(0) = \sum_\mu c_\mu(0)\mathbf{v}_\mu$ into Eq. (4) gives $\mathbf{R}(t) = A(t)\sum_\mu c_\mu(0)e^{-\kappa_\mu t/\tau}\mathbf{v}_\mu$. The first term contains the first eigenvector $\mathbf{v}_1 = \mathbf{1}$ and corresponds to the fixed point of the simpler model $\mathbf{R}(t) = P(t)\mathbf{R}(0)$ (that is, the model with the factor $A(t)$ fixed at 1). Separating out the first term from others, we have

$$\mathbf{R}(t) = A(t)\left(c_1(0)\mathbf{1} + \sum_{\mu \geq 2} c_\mu(0)e^{-t/\tau_\mu}\mathbf{v}_\mu\right), \tag{5}$$

where we defined $\tau_\mu \equiv \tau/\kappa_\mu$.

Equation (5) describes the following behavior. First, neglecting the effect of the shift factor $A(t)$, the activity basket of a region converges over time to a uniform vector $c_1(0)\mathbf{1}$. In this state, the region has equal ability in all products. On its way to this state, the basket shows higher ability in some products and lower ability in others, with each eigenvector $\mathbf{v}_\mu$ describing a different pattern of deviations from the long-term steady state of uniform ability across products. Each eigenvector $\mathbf{v}_\mu$ ($\mu \geq 2$) satisfies $\mathbf{w}_1^\dagger \mathbf{v}_\mu = 0$ where $\mathbf{w}_1$ has only positive elements, and so each such eigenvector contains a mix of positive and negative elements. Each pattern of deviation decays with time at the rate set by τ_μ, the characteristic time scale of the μth eigenmode. If the pre-factor $A(t)$ were fixed and equal to 1, a region would be destined to have ability $c_1(0)$ in every product. Letting $A(t)$ change, the final ability in products can be arbitrarily high or low.

To understand the implications of the PoR for long-term economic change, we focus on the two eigenmodes with the longest time scales. Absorbing the coefficient $c_1(0)$ into $A(t)$, and defining the coefficient $b(t) \equiv c_2(0)e^{-t/\tau_2}$, the longest-lived dynamics of $\mathbf{R}(t)$ are described by the two leading terms of Eq. (5):

$$\mathbf{R}(t) \sim A(t)\mathbf{1} + A(t)b(t)\mathbf{v}_2. \tag{6}$$

The coefficients $A(t)$ and $b(t)$ have simple interpretations. A shift in $A(t)$ corresponds to a region realizing a uniform change in abilities across activities. A shift in $b(t)$ corresponds to a compositional shift. Some activities rise in ability and others fall, as determined by the signs and magnitudes of entries in $\mathbf{v}_2$. Together, these coordinates situate the activity basket $\mathbf{R}(t)$ in a 2D space.

To further interpret these coordinates, let $\boldsymbol{\pi} = \mathbf{w}_1$ denote the first left eigenvector of L_Φ, normalized so that its elements sum to 1. By again exploiting the biorthogonality of the eigenvectors one can show

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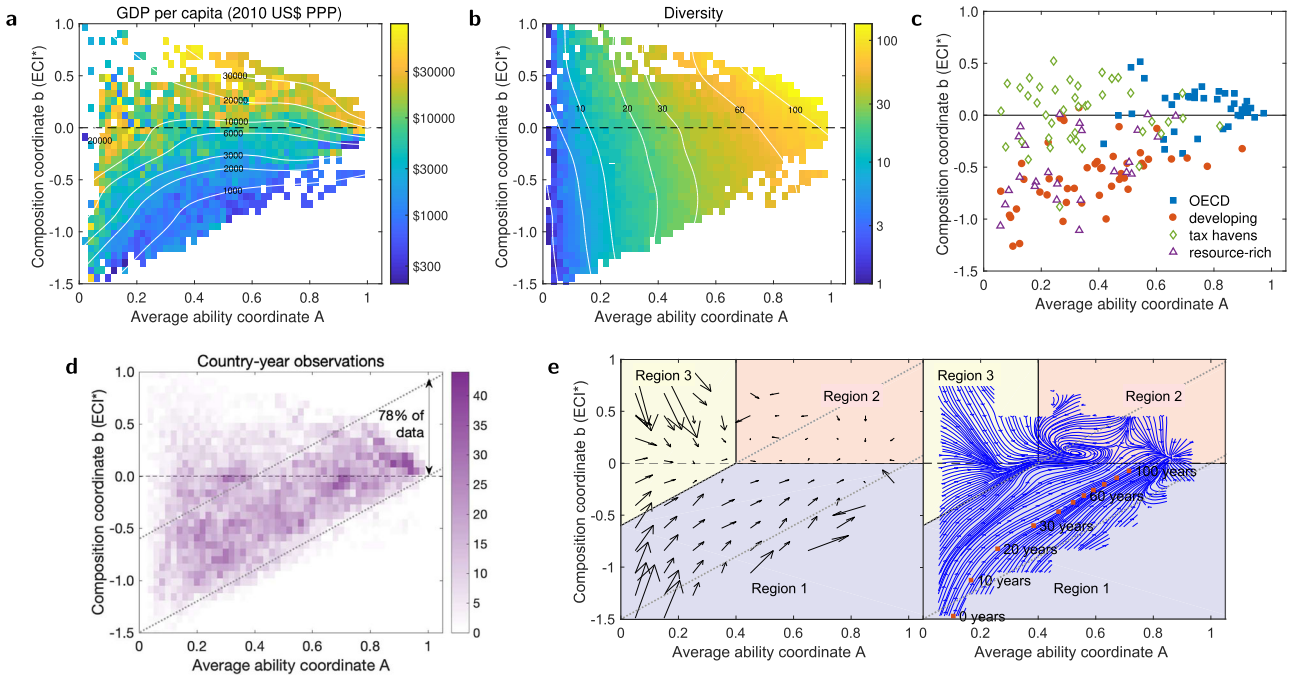

Fig. 3 | Country export baskets plotted according to average ability (*A* coordinate) and composition (*b* coordinate). Plots are based on Comtrade data[47] for 249 countries and regions over the period 1962–2018 (11,544 observations). We analyze these data at the 3-digit Standard International Trade Classification (SITC) product level (235 product categories), and obtain similar results using 3-digit Harmonized System (HS) codes (Supplementary Note 2). See Supplementary Note 3 for discussion of trajectories and locations of individual countries. **a**, **b** We form bins in the 2D plane to collect country-year observations, and compute average GDP per capita and average diversity d within each bin. White lines are contours of a LOWESS-smoothed surface fit to bin averages. **c** Positions of export baskets in 2018 of four categories of economies. 'Developing' refers to economies on the UN's least developed economies list[69], 'tax havens' to non-OECD and non-developing countries appearing in Hines (2010)[48], and `resource-rich' to economies not elsewhere categorized whose natural resource rent[70] exceeded 5% of GDP in 2018. **d** Number of country-year observations in each bin. **e** Quiver and streamline plots for the average directional change over 20 years for countries beginning at different starting points in the plane. Red dots in the right panel follow one streamline and roughly convey the average pace of movement, though no streamline should be taken as a typical country trajectory; individual countries display wide variation in both direction and speed of change. To construct the quiver plot, we divided the plane into 10 equal-sized bins horizontally and 12 equal-sized bins vertically (120 bins total). For each bin, we obtain the horizontal and vertical components of movement by computing the average horizontal change ΔA and average vertical change Δb over the next 20 years using all observations starting in the bin. We used the MATLAB quiver function to render a vector field and the streamslice function to render a streamline plot.

("Methods", Eq. (14)) that

$$A(t) = \boldsymbol{\pi}^T \mathbf{R}(t) \equiv \bar{R}(t). \tag{7}$$

That is, $A(t)$ is a weighted average $\bar{R}(t) \equiv \boldsymbol{\pi}^T \mathbf{R}(t)$ of the abilities of a region. For this reason, we refer to $A(t)$ as the average ability coordinate. To further interpret $b(t)$ we can write Eq. (6) as $\mathbf{R}(t) \sim A(t)\left[\mathbf{1} + b(t)\mathbf{v}_2\right]$ and observe that $b(t)$ only characterizes the deviation of $\mathbf{R}(t)$ from uniformity, independent of the average level of these abilities that is captured by $A(t)$.

Diversity and composition of exports baskets over time

The PoR thus delivers coordinates that theoretically could be used to characterize the activity basket of an economy. We now examine these coordinates empirically, asking whether their movements corroborate facts of economic development. We focus on international trade data from the United Nations (UN) Comtrade database[47]. These data are frequently used in research on economic complexity because they offer a detailed description of what economies export over long periods of time. We use our coordinates to describe the evolution of export baskets from 249 countries over the 57-year period 1962–2018. This exercise summarizes country development by two simple statistics.

RCAs are commonly transformed to mitigate the influence of extreme values, and here we transform the heavy-tailed RCAs with the function $g(R) \sim \log(1 + R/R_0)$. We tune the parameter R_0 to maximize the variance that the b coordinate can explain across time and countries (Methods, "Transformation of RCAs"). We then calculate for each country c and time t the RCA vector $\mathbf{R}_c(t)$ on this transformed scale and use these vectors to compute the A and b coordinates.

Figure 3a, b depicts country income and export diversity as functions of A and b. This lets us observe how each quantity varies as a function of one coordinate while holding the other fixed. For reasons that will become clear later when we compare our structural change coordinates with complexity metrics, we also refer to b as ECI* in these graphs.

Income increases with *b*, diversity with *A*. A higher value of the compositional coordinate b for a country's export basket is associated with significantly higher Gross Domestic Product (GDP) per capita (Fig. 3a). For example, holding A fixed at 0.5, an increase in b from −0.5 to +0.5 is associated with an increase in a country's per capita income from about $1500 to $30,000 (2010 US dollars at Purchasing Power Parity or PPP). In contrast, the association between a country's average ability A and its income when holding the value of b fixed is weak at best.

The situation is roughly opposite when we examine countries' export diversity (Fig. 3b). We measure diversity as the count of products d_c in which a region c has an RCA greater than 1. Higher values of the average ability coordinate A are closely associated with greater diversity in a country's export basket. Holding b fixed at 0, an increase in A from 0.1 to 0.9 is associated with an over 10-fold rise in the number of products with RCA above 1. In contrast, when holding A fixed, a

compositional shift towards higher levels of b is only weakly associated with higher export diversity.

Different types of countries inhabit different regions of this coordinate system (Fig. 3c). The developed countries of the Organization for Economic Co-operation and Development (OECD) are predominantly located in the right of the plot with relatively high values of both A and b, corresponding to diverse, high-income, industrialized economies. In the lower left, we find a set of undiversified developing economies as defined by the UN's least developed economies classification. In contrast, in the upper left, we find countries that are also undiversified, yet frequently have high-income. Many of these are resource-rich economies or are often considered tax havens[48].

Notably, countries often diversify as they rise in income[32], but the observations above do not directly associate higher income with higher diversity of exported products. Rather, they suggest that what matters more for a country's GDP per capita is the composition of exports[1]. We next explore the dynamics of countries in this space and further probe this distinction.

Country development. A particularly densely occupied portion of the A-b plane is a diagonal band that stretches from low A / low b to high A / above-average b (Fig. 3d). We first focus on the movement of countries whose export baskets lie in this main band. Countries show diverse trajectories (see Supplementary Note 3). To characterize broad tendencies over long periods, we group country-year observations into bins in the A-b plane and examine the average direction and speed of movement over the subsequent 20 years for observations that start within a given bin. We summarize our results by referring to three regions of the plane, labeled Regions 1, 2, and 3 (Fig. 3e).

Countries beginning in the lower part of the main diagonal band (Region 1) trend over decadal time scales upward and to the right. Given how A and b are defined, a simultaneous increase in both coordinates corresponds to an export basket simultaneously realizing (1) a general improvement in ability across products, and (2) a shift in composition towards products with positive values in the vector $\mathbf{v}_2$. These two components of movement are associated with different effects. The shift in the composition of exports captured by b is (empirically) most directly associated with increased GDP per capita (Fig. 3a), while the increase in A is more associated with a rise in export diversity (Fig. 3b). These patterns are consistent with the idea that higher income is associated with exporting particular products, rather than diversification per se. Nevertheless, in practice, countries in the lower part of the main diagonal band that succeed in reaching such products tend to do so while diversifying into a broad range of goods, including ones not associated with higher income.

What products are involved in the vertical shifts that increase b? Manufacturing products very often have positive elements in $\mathbf{v}_2$, and agricultural products very often have negative elements. As a result, countries in Region 1 that traverse the length of the diagonal band see a broad, long-lived shift in their export baskets away from agricultural products and towards manufactures. This shift is consistent with the long-observed tendency for economies to move from agriculture to manufacturing (and then on to services, a move that largely eludes trade statistics) as they develop[31,34].

If a country traverses the length of the diagonal band in Region 1, it will arrive in Region 2, where countries have high-income, diverse, developed economies. Within this region, the average speed of movement is much lower than in Region 1. This in part owes to the fact that export baskets here evolve in a greater variety of directions, with an average directional change near zero. Broadly, though, countries in Region 2 tend to sustain a high b, somewhat above zero, and move within a range of relatively high A values. On reaching Region 2, a number of countries move toward lower values of A. This transformation path is consistent with a phenomenon in which countries see a fall in diversity in late stages of development[32,33].

Finally, countries in Region 3 tend to have high income and low diversity. Many of these countries are abundant in natural resources (particularly oil) or function as tax havens. In this region, countries tend to move quickly toward lower levels of b, converging near zero. We are not aware of any prior observations that this movement corresponds to. Evidently, it is difficult for these countries to sustain a high degree of specialization in products that load positively on $\mathbf{v}_2$ for long periods of time.

Together, these observations show an agreement in our framework between short- and long-term economic change. The Principle of Relatedness describes structural change on short time-scales and fine-grained levels of sectoral resolution. When we analyze network models that operationalize this principle, using the standard technique of eigenmode decomposition, we arrive at coordinates that should capture activity changes over long time-scales and higher levels of aggregation. As one would hope, observed movements of export baskets in these coordinates yield long-term and coarse-grained descriptions of structural change that are consistent with well-documented stylized facts.

Three further remarks are in order. First, higher-order eigenvectors beyond the second are of interest because, in principle, they could describe other important modes of transformation. However, the proximity matrices in the literature only show strong agreement in the structure of the first two eigenvectors, and do not straightforwardly resolve how many eigenvectors matter (see Supplementary Note 4). Second, note that these directional vectors are based on data from just 1962 (the first available year in our dataset). As is typical in studies of the PoR, we build a network of activities from one year of data and use it to predict subsequent changes in countries' exports. In later years, we are observing where countries lie on axes defined by the 1962 network. This procedure could be viewed as a prediction of country evolution (where we assume countries evolve according to a simple spreading model on a static network defined by past data), or as a way to empirically describe country evolution in a particular coordinate system (one that benchmarks countries against a hypothetical country obeying the PoR). Second, since the PoR is a heavily stylized description of country evolution, it is notable that the PoR is sufficient to reproduce key features of the canonical development path. Nevertheless, countries also violate the PoR, such as by making jumps into unrelated industries (see, for instance[49,50]). Modeling this and other mechanisms of change could, for example, mean adding terms to Eq. (2). Changing the model will, in turn, modify its long-run predictions and the preferred coordinate system in which to describe change, potentially improving these predictions or yielding better quantities to summarize the evolution of country activities.

Comparing our structural change coordinates with complexity metrics

Having seen that the PoR leads to coordinates that track structural change, we now show how they also closely resemble economic complexity metrics, even though the latter are motivated very differently than the coordinates here. Complexity metrics have been put forward as tools to infer the number of production capabilities an economy possesses based on observations of which activities are performed in which places. Proposals include the Economic Complexity Index (ECI)[18], country Fitness[19,23], the entropic measure of Teza, Caraglio, and Stella[24,29], Production Ability[26], GENEPY[27], and collective knowhow[30]. We first describe the relationship between these metrics and our coordinates and then comment on its significance.

Different complexity metrics are obtained through a wide range of theoretical derivations, but empirically, they fall into two main groups (Fig. 1) with a straightforward correspondence to the coordinates that emerge from the PoR. The first group contains diversity-like quantities. These include diversity d_c itself, the A coordinate of our framework, country Fitness, Production Ability, and the entropic measure of Teza, Caraglio, and Stella. The other group contains

quantities that capture a particular type of compositional information about an economy; roughly, the agriculture-to-manufacturing axis noted earlier. This group includes the ECI, and the b coordinate of our framework (ECI*). The GENEPY metric by design combines information associated with both groups, and itself does not fall clearly into either one, but its first component X_1 by construction is related to Fitness and is associated with the first group, while its second component X_2 by construction is related to the ECI and is associated with the second group. (See Supplementary Note 5 for further discussion of complexity metric correlations).

These correlations can be anticipated on theoretical grounds. The diversity-like quantity $A = \sum_p R_{cp}\pi_p$ we derive is a sum over activities that a country performs with significant ability, weighted by the all-positive elements π_p. Similarly, the Fitness metric $F_c = \sum_p M_{cp}Q_p$ is a sum over the activities that a country performs with significant ability, with weights given by the all-positive product Qualities Q_p. Not surprisingly, these quantities strongly correlate with each other and with the count of products in which a country has an RCA greater than 1, $d_c = \sum_p M_{cp}$, a standard measure of diversity.

Similarly, the $b(t)$ coordinate has a close theoretical connection to the ECI and its counterpart metric, the Product Complexity Index or PCI. The PCI and ECI were proposed by Hidalgo and Hausmann (2009)[18], and can be computed with an eigenvector computation as noted by Caldarelli et al. (2012)[51,52]. When proximities are measured as $\Phi^P_{pp'} \equiv \sum_c \frac{M_{cp}M_{cp'}}{D_c}$, the second vector $\mathbf{v}_2$ of the dynamical model and the vector of PCIs (**PCI**) solve the same eigenvector equation ("Methods", "The eigenvector $\mathbf{v}_2$ (PCI*)"), and are therefore identical up to a normalizing constant. There is ambiguity about which proximity measure one should use to build the network of activities, but we find that the second eigenvector is not sensitive to this choice (Supplementary Note 6) and consistently has very high correlation with **PCI**. To distinguish the conventional PCI vector and the second eigenvector of the dynamical model, we call the latter $\mathbf{PCI}^* \equiv \mathbf{v}_2$. Thus, in general, $\mathbf{PCI}^*$ and **PCI** are strongly correlated, and in the special case $\Phi = \Phi^P$, they are identical. The $b(t)$ coordinate that tracks where a country lies on the axis of change described by $\mathbf{v}_2 = \mathbf{PCI}^*$ is in turn, closely related to the ECI. We define an ECI-like quantity in our framework, $\mathrm{ECI}^*(t) \equiv b(t)$. To see that this quantity is closely related to the ECI, note that the ECI equals the average PCI of the products in which a country has an RCA greater than 1:[53,54]

$$\mathrm{ECI}_c = \sum_p \left(\frac{M_{cp}}{\sum_{p'} M_{cp'}} \right) \mathrm{PCI}_p. \tag{8}$$

The ECI* in turn can be computed by observing that $b(t) = c_2(t)/c_1(t)$ and calculating the eigen-expansion coefficients c_2 and c_1. This calculation ("Methods", "The b coordinate (ECI*)") gives

$$\mathrm{ECI}^*_c = \sum_p \left(\frac{R_{cp}\pi_p}{\sum_{p'} R_{cp'}\pi_{p'}} \right) \mathrm{PCI}^*_p. \tag{9}$$

The ECI and $b(t)$ are just different averages of PCIs or PCI*s. The conventional ECI gives equal weight to activities in which the country has an RCA above 1, and zero weight to other activities. The ECI* weights activities by RCA, adjusted by the elements of $\boldsymbol{\pi} = \mathbf{w}_1$. Not surprisingly, these averages are strongly correlated (Fig. 4), and this holds for any proximity matrices we consider (Supplementary Note 6). (See Supplementary Note 7 for rankings of countries by ECI*).

Interpretation. What do we make of the resemblance of complexity metrics to the structural change coordinates that emerge from the PoR? First, it demonstrates (echoing and expanding on Ref. 53) that quantities very similar to complexity metrics can be motivated with arguments that have little, per se, to do with 'complexity'. The relevance of $A(t)$ and $b(t)$ is not tied to considerations of how to infer hidden information about capabilities, but to how aptly they characterize overt long-lived patterns of change in economies (Table 1).

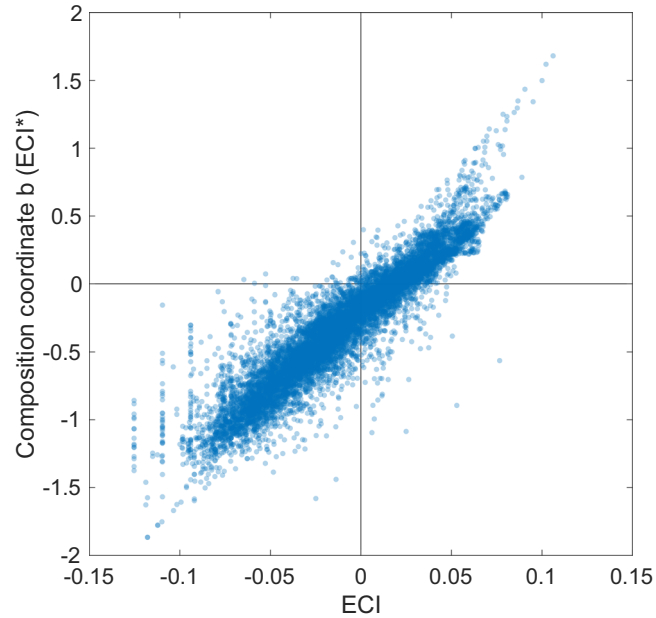

Fig. 4 | The *b* coordinate (ECI*) versus ECI. Sample consists of country-year combinations that cover 249 countries and regions and 57 years. The ECI* and ECI are computed from Comtrade data[47]. The ECI is the standard Economic Complexity Index of [18], which is motivated as an inference method to track the complexity of an economy in terms of latent capabilities that underly its export mix. The ECI* instead represents the b-coordinate in our long-run structural transformation plane, expressing one of the two principle axes of long-run structural transformation implicit in the short-run dynamics of the Principle or Relatedness. The alignment between the two metrics shows that measures that strongly resemble complexity metrics can be motivated without reference to capabilities or complexity and instead by expressing a country's progress along an axis of particularly long-lasting structural change. Here, the ECI and ECI* represent differently weighted averages of the same PCI* vector (the second eigenvector of the product proximity matrix). In Supplementary Note 6, we show that the strong resemblance between ECI and ECI* also persists if we use different product proximity matrices to capture the short-run Principle of Relatedness dynamics (with different resulting PCI* vectors).

Second, our findings suggest debates between the main contending metrics could be misguided. Complexity metrics are typically viewed as being in competition, since they represent alternative methods to estimate the same quantity (country or product complexity). But clearly, we would not view the coordinates $A(t)$ and $b(t)$ this way; these coordinates just summarize different and overt information about activity baskets, focused either on their diversity ($A(t)$) or composition ($b(t)$). The results here affirm the empirical relevance of complexity metrics while raising the question of whether they infer complexity or just summarize development-related changes in an economy's basket of activities. In the latter case, different metrics would emphasize changes to either the diversity or composition of activities, but—as Fig. 3 illustrates and classic and recent literature supports[31,32,34,35]—both changes are general features of development.

Third, despite similarities of our coordinates to prominent complexity metrics, our framework comes with a different structure and intuition. Several complexity metrics emerge from a framework in which country and product metrics are co-determined by a system of equations[18,19,27]:

$$\text{country complexity} = F(\text{product complexities}) \tag{10}$$

$$\text{product complexity} = G(\text{country complexities}). \tag{11}$$

Table 1 | Comparing the usual view of complexity metrics and the dynamical systems view here, based on the Principle of Relatedness

	Complexity inference view	Dynamical systems view
Goal	Solve an inference problem: Estimate country and product complexities.	Characterize activity baskets: Describe baskets of economic activities in low-order terms using particular coordinates.
Competitors or complements?	Different metrics are in competition. They offer different methods for solving the inference problem above.	Metrics belonging to different classes in Fig. 1 are largely complementary. They summarize different information (diversity, composition) about an economy.
Mathematical derivation	Motivated by a data challenge: Infer complexity from data about what goods are produced where.	Derived from a model: Coordinates emerge from a dynamical model of economic diversification.
Role for Principle of Relatedness?	None in particular.	Yes—the Principle of Relatedness implies the relevance of particular coordinates.
Product and country metrics...	...have similar interpretations: Both measure complexity. In some setups, these metrics solve a set of simultaneous equations.	...have different interpretations: Product metrics collectively describe a direction of economic change, country metrics individually show where a country lies along this axis. Product metrics are computed first, and country metrics follow.

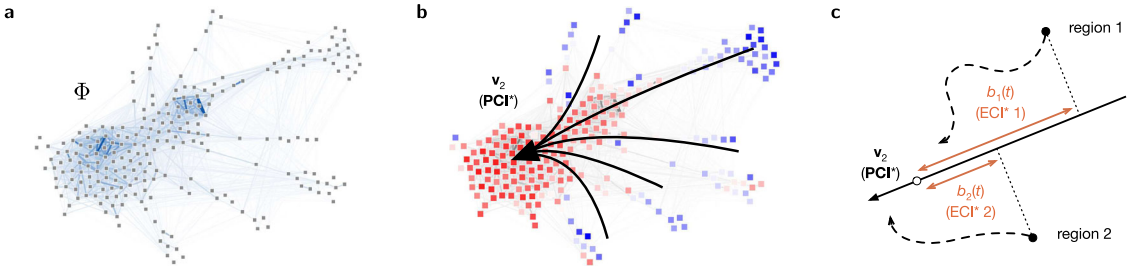

Fig. 5 | Order of computation of the framework's key quantities. a A relatedness network is defined that captures similarities among products. This product-similarity matrix, Φ, defines the short-term diversification trajectories of countries. These short-run dynamics are known as the Principle of Relatedness: economies' tendency to preferentially diversify and grow products that are closely related to the ones they currently produce, while abandoning or down-sizing products that are unrelated to their current export baskets. **b** The long-term dynamics of the Principle of Relatedness can be approximated by the compositional mode of change described by the elements of $\mathbf{v}_2$ (**PCI***). Negative values (blue) correspond to economic activities that are vacated in relative terms, while positive values (red) correspond to activities that are reached. **c** A region's export basket can be projected onto this mode. The country's progression in this projection along the PCI* coordinate determines its value of b (ECI*). Analogous projections and dynamics can be displayed for the progression along the A-coordinate. Together, the A and b-coordinates define a country's position in the long-term transformation trajectory implied by the short-run dynamics described by the Principle of Relatedness.

Different metrics arise from different choices for functions F and G. For example, one obtains the ECI by taking F to be the arithmetic average complexity of the products that a place exports competitively (that is, with RCA > 1), and G to be the arithmetic average complexity of the countries that competitively export a given product. Alternatively, one obtains country Fitness by replacing G with a harmonic average. In contrast, here there is no co-determination but a two-step sequence (Fig. 5a–c): The eigenvectors of the relatedness network are first computed, describing various directions of change, followed by a region's coordinate projections onto these axes. Relatedly, several product metrics[18,27] are derived from a product similarity matrix, while country metrics are derived from a country similarity matrix. In contrast, here we use just one similarity matrix—between products. Under the mindset we adopt to derive our results, this matrix is fundamental, capturing technological relationships between products, whereas the country similarity matrix is incidental, capturing current similarities between countries depending on their current stage of development. Our asymmetric treatment of countries and products is underscored by the fact that country and product quantities can be estimated with different data. In our framework, countries' coordinates could be computed with trade data, while the product proximity matrix is constructed from other data. (See Supplementary Note 8 for further discussion).

Fourth, our results would help make sense of conundrums that have sometimes been swept under the rug. For example, oil production is ranked near the bottom of complexity by both the PCI and product Quality measures—yet oil production is clearly a complex activity, requiring knowledge of multiple fields of engineering, geology, chemistry, physics, transportation and logistics, business operations, and other areas. At the other end of the spectrum, pottery and works of art are ranked highly by these metrics, yet are plausibly low in complexity. Direct measures of complexity based on occupational diversity (Supplementary Note 9) are also in tension with the complexity rankings obtained from these metrics. However, these cases are not mysterious if these metrics do not infer complexity, but track structural change. Then these rankings would simply reflect the fact that oil is a product that economies diversify away from as they develop, and that pottery and artwork are activities most readily supported in rich and developed economies. (Explanations for the "resource curse" have long been debated, see for instance[55]).

Our results suggest that prominent complexity metrics summarize structural change rather than infer hidden information about capabilities. However, it is possible that, under certain assumptions about the distribution of comparative advantage, these metrics closely resemble such inference metrics. Recent works explore this by making additional assumptions about the structure of capabilities and the development process[25,26,30]. Accordingly, a correspondence between complexity and long-term change is likely and would reflect the fact that economies become more complex over time. However, why and how would have to be made clear. (Presumably, the goal of development is raising productivity or well-being, not complexity per se).

Dimensionality reduction. We now discuss several works that approach economic complexity from the perspective of dimensionality reduction. First, note our method of analysis (eigenmode decomposition) is a dimensionality-reduction technique—it returns axes (the

eigenvectors $\mathbf{v}_\mu$) on which one can describe high-dimensional data (the activity baskets $\mathbf{R}(t)$) and the coordinate projections of such data onto these axes (the coefficients $c_\mu(t)$). Differences between techniques arise in how directions in the data space are determined and interpreted, with dimensionality-reduction methods typically aiming to identify directions that capture high amounts of variation in data, while eigenmodes specifically characterize coherent patterns in dynamics. In fact, an eigenmode decomposition not only generates a representation of data in a low-dimensional vector space, but also shows that the model predicts that countries' activity baskets will converge to such a subspace (spanned by the directions with the largest time scales). Differences between countries shrink first among quickly-decaying dimensions, leaving activity baskets scattered primarily along slowly-decaying ones, a well-known outcome of the interaction of structure and dynamics in networks[44,56] (Supplementary Note 10).

Mealy et al. (2019)[53] point out that the conventional ECI is equivalent to spectral clustering, ordering regions along an axis that scores the similarity in their baskets of activities. Our results are consistent with this view and extend it, showing that the ECI is not just a similarity score but also tracks a long-lived pattern of compositional change. In addition, our results relate not just to the ECI but also to diversity-like complexity metrics. Sciarra et al. (2020)[27] put forward the GENEPY metric that reduces the dimensionality of economic activity data by combining information from two eigenvectors of a country similarity graph. This resembles what we do in that we also exploit a 2-D picture of country development, but there are technical and conceptual differences. The $X_{c,1}$ and $X_{c,2}$ components underlying GENEPY are elements of eigenvectors of a country similarity graph, while we have no such graph; our coordinates come from projecting export baskets onto the eigenvectors of a product similarity graph. It is also not $X_{c,1}$ and $X_{c,2}$ themselves that correspond to our axes, but transformations of these components involving diversity d (Supplementary Note 5). Conceptually, GENEPY is derived with the assumption that the ECI and country Fitness both represent (imperfect) measures of complexity, and can potentially be improved by combining them into the hybrid GENEPY measure. But as we emphasize, our coordinates can just be understood as projections onto different axes of change, with straightforward interpretations in terms of development. Finally, Brummitt et al. (2020)[35] apply machine learning to historical export data to extract dimensions that characterize variation in export baskets across countries and time, finding axes that strongly correlate with the PCIs. This is closely related to the work here, because the first principal component in Brummitt et al. captures a simultaneous increase in export diversity and a compositional shift, precisely the two kinds of changes that our analytically-derived coordinates separate.

Prediction of income and growth. Complexity metrics have been justified in part by their empirical connection to country income and economic growth, and the literature often evaluates different metrics by how well they predict these variables. For completeness, we examine this ability for the A and b coordinates (Supplementary Note 11), finding that the b coordinate predicts income and growth better than the conventional ECI, country Fitness, and GENEPY. This finding is interesting, though it is unclear what should be concluded from it. Apparently, growth rates are connected to the long-run compositional shifts predicted by the PoR, but this is not empirically surprising, and our dynamical model makes no assumptions about how a country's activity basket relates to its level of wealth. Ultimately, understanding how and why the PoR entails a shift to baskets of activities that are more growth-enhancing than others is an open question.

Discussion

The two branches of economic complexity – the Principle of Relatedness and complexity indices—have been influential in academic and policy debates about economic development. However, there is confusion about the formal relationship between these branches and debate on how to interpret some of the field's results. Economic complexity builds on a narrative centered on capabilities, with complexity metrics intended as data inference methods under this narrative, but the mathematical linkage between metrics and narrative is not tight. In addition, while the PoR and complexity metrics represent new approaches to understanding structural change, their relationship to classic understandings of structural change is underdeveloped. Our work sets up such a relationship, in the process adding granularity and nuance to established facts of economic development. Our results also suggest that prominent complexity metrics describe overt changes in activities that economies undergo, rather than infer information about capabilities. Such changes in how we view complexity metrics may have a bearing on their use in policy settings.

There are two broad ways to view our results. First, we developed a modeling framework that connects short- and long-term descriptions of economic structural change. The Principle of Relatedness describes structural change on short, year-to-year time scales and across fine-grained activities, while canonical observations[31,34] emphasize multi-decadal transitions across broad sectors as an economy diversifies[32]. As one would hope, a modeling framework based on short-run observations is consistent with the long-run and extends long-run observations, describing them in granular terms. This elaborates on a long-standing goal, dating back at least to Kuznets[31], to understand trajectories of economic transformation. Second, our results tie together two areas of work in the rapidly growing field of economic complexity. The field's major branches—studies of relatedness and complexity metrics—share concepts and motivating questions, but are linked more in spirit than in math. We connect these branches by showing mathematically and empirically how complexity metrics track two kinds of long-term transformations that are implied by PoR models. This suggests these two branches describe short-term and long-term consequences of the same assumptions about economic development.

Recent work has harnessed new concepts and analytical tools to probe economic development. Numerous avenues for exploration remain, of which we mention two. First, we explored our results in the setting of country exports, but similar analyses could be carried out in networks of occupations[57], industries[9,16,58], technology classes[59], research publications[5], or in networks of related locations[60], where the roles of locations and activities are reversed by projecting the bipartite network of locations and activities onto locations instead of activity nodes. Second, while it is valuable to see that the canonical development pathway emerges just from assuming a country obeys the PoR, the PoR is a highly simplified model that leaves out numerous processes of change (for example, long-distance leaps to unrelated industries[49]). Our results call for improving the dynamical description of structural change. Describing such processes with greater fidelity may shed further light on the determinants of growth and development.

Methods

The A coordinate

The $A(t)$ factor in Eq. (4) can be understood as a weighted average of the elements of $\mathbf{R}(t)$. To see this, consider the simpler model $\mathbf{R}(t) = P(t)\mathbf{R}(0)$ in which $A(t)$ is fixed at 1. Left-multiplying by the first left eigenvector $\mathbf{w}_1 = \boldsymbol{\pi}$ (normalized so that its elements sum to 1), and exploiting the fact that $\boldsymbol{\pi}$ is the stationary state of the matrix $P(t) = e^{-L_\Phi(t/\tau)}$, we have

$$\bar{R}(t) = \boldsymbol{\pi}^T\mathbf{R}(t) = \boldsymbol{\pi}^T\mathbf{R}(0) = \bar{R}(0). \tag{12}$$

This demonstrates a well-known aspect of consensus dynamics models (of which $\mathbf{R}(t) = P(t)\mathbf{R}(0)$ is an instance): The dynamics preserves the average value $\bar{R}(0) = \boldsymbol{\pi}^T\mathbf{R}(0)$ of the initial condition $\mathbf{R}(0)$ (see, for instance[44]).

If we then allow $A(t)$ to vary in Eq. (4), it is clear that the average of $\mathbf{R}$ will change by whatever factor $A(t)$ changes. Left-multiplying Eq. (4)

by $\boldsymbol{\pi}$, after rearrangement, we have

$$A(t) = \frac{\boldsymbol{\pi}^T \mathbf{R}(t)}{\boldsymbol{\pi}^T \mathbf{R}(0)} = \frac{\bar{R}(t)}{\bar{R}(0)}, \tag{13}$$

showing that $A(t)$ is the factor by which the average of $\mathbf{R}$ changed between times 0 and t. The time $t = 0$ is just a reference period, with no special significance, and $\bar{R}(0)$ just a reference value. Taking this value to be 1, we have

$$A(t) = \boldsymbol{\pi}^T \mathbf{R}(t) = \bar{R}(t). \tag{14}$$

The same result can also be obtained from the eigen-expansion Eq. (5). Left-multiplying by $\mathbf{w}_1 = \boldsymbol{\pi}$, and exploiting the biorthogonality of left and right eigenvectors, we have $\boldsymbol{\pi}^T \mathbf{R}(t) = A(t) c_1(0) \boldsymbol{\pi}^T \mathbf{1} = A(t) c_1(0)$. The $c_1(0)$ coefficient plays a redundant role with $A(t)$; setting it to 1 leaves us with Eq. (14).

Review of the Product Complexity Index and Economic Complexity Index

The PCI and the ECI were proposed by Hidalgo and Hausmann (2009)[18], and can be computed with an eigenvector computation as noted by Caldarelli et al. (2012)[51,52]. Let n_c be the number of countries and n_p the number of products. Let R be the $n_c \times n_p$ matrix R of RCAs, and let M be a binarized version of this matrix, with elements $M_{cp} = 1$ wherever $R_{cp} \geq 1$ and 0 otherwise. M may be viewed as an adjacency matrix for a bipartite network connecting countries to products they export with significant ability. Define the diversity of country c's activity basket as the cth row sum of this matrix, denoted $d_c \equiv \sum_p M_{cp}$. Similarly, define the ubiquity of activity p as the pth column sum, denoted $u_p \equiv \sum_c M_{cp}$. Let D and U be the diagonal matrices formed from countries' diversities and products' ubiquities, respectively. One may use the products of $D^{-1}M$ and $U^{-1}M^T$, in different orders, to define two square matrices that eliminate either the country nodes or the product nodes from the bipartite network. Multiplying these matrices in one order collapses the country dimension, leading to the $n_p \times n_p$ row-normalized stochastic matrix

$$\mathcal{P} \equiv U^{-1}(M^T D^{-1} M). \tag{15}$$

The second right eigenvector of $\mathcal{P}$ defines the unstandardized PCIs, **PCI**. Similarly, collapsing the product dimension leads to the $n_c \times n_c$ row-normalized stochastic matrix

$$\mathcal{C} \equiv D^{-1}(M U^{-1} M^T). \tag{16}$$

The second right eigenvector of $\mathcal{C}$ defines the unstandardized ECIs, **ECI**. Often, the PCIs and ECIs are standardized to have zero mean and unit variance across products or countries[61], but here we work with the unstandardized vectors, and indeed our theory implies that standardizing removes information (see Supplementary Note 12).

The eigenvector $\mathbf{v}_2$ (PCI*)

The second eigenvector of the Laplacian matrix of the dynamical model $\mathbf{v}_2$ captures the pattern of compositional change with the longest time scale. This mode of change represents a relative shift from some activities to others as determined by the signs of the elements of $\mathbf{v}_2$, giving this vector meaning as a pattern of economic transformation. We develop here how the elements of $\mathbf{v}_2$ are closely related to the complexity metrics known as the Product Complexity Indices. To highlight its relation to the PCIs, we refer to this eigenvector as $\mathbf{PCI}^*$:

$$\mathbf{PCI}^* \equiv \mathbf{v}_2. \tag{17}$$

To see that elements of $\mathbf{v}_2$ are related to PCIs, first note that the eigenvectors of our dynamical model depend on how we construct the matrix of proximities between activities. Prior work has generated a variety of proximity matrices, each finding empirical support when used to forecast transitions in economic activities. It is not clear what proximity measure describes the network of transitions best, and our goal here is not to resolve this. Instead, we show that the structure of the second eigenvector is not sensitive to this choice, and that across a variety of empirically-supported measures of proximity (Supplementary Note 6), the second eigenvectors are quite similar to one another, and to the vector of the conventional PCIs.

First, there is a particular choice of proximity matrix for which $\mathbf{v}_2 = \mathbf{PCI}^*$ and the vector of conventional PCIs, **PCI**, is identical up to an irrelevant factor. This happens when the proximities are given by

$$\Phi^P_{pp'} \equiv \sum_c \frac{M_{cp} M_{cp'}}{D_c}, \tag{18}$$

or in matrix form $\Phi^P = M^T D^{-1} M$. If the proximities Φ used in the dynamical model are given by Φ^P, then $\mathbf{PCI}^*$ and **PCI** will be the second eigenvectors of the same eigenvector equation. To see this, note that $\mathbf{PCI}^*$ is by definition an eigenvector of $L_\Phi = I - \tilde{\Phi}$ and therefore an eigenvector of $\tilde{\Phi}$ as well. It holds that $\tilde{\Phi} \equiv \mathrm{Diag}(\Phi \mathbf{1})^{-1} \Phi$ equals the matrix $\mathcal{P}$ whose second eigenvector defines the conventional PCIs:

$$\begin{aligned}
\tilde{\Phi} &\equiv \mathrm{Diag}(\Phi^P \mathbf{1})^{-1} \Phi^P \\
&= \mathrm{Diag}(M^T D^{-1} M \mathbf{1})^{-1} M^T D^{-1} M \\
&= \mathrm{Diag}(M^T D^{-1} \mathbf{d})^{-1} M^T D^{-1} M \\
&= \mathrm{Diag}(M^T \mathbf{1})^{-1} M^T D^{-1} M \\
&= \mathrm{Diag}(\mathbf{u})^{-1} M^T D^{-1} M \\
&= U^{-1} M^T D^{-1} M \\
&= \mathcal{P}.
\end{aligned} \tag{19}$$

Further, a close numerical relation between $\mathbf{PCI}^*$ and **PCI** carries over beyond this special case; see Supplementary Note 6.

The b coordinate (ECI*)

The b coordinate is related to the projection of a country's ability vector onto $\mathbf{v}_2$. This coordinate can be written in the form of Eq. (9), showing that it closely resembles the conventional ECI. To derive Eq. (9) we left-multiply $\mathbf{R}(t) = \sum_\mu c_\mu(t) \mathbf{v}_\mu$ by $\mathbf{w}^\dagger_\mu$ and exploit the biorthogonality of the left and right eigenvectors to obtain $c_\mu(t) = \mathbf{w}^\dagger_\mu \mathbf{R}(t)$. Examining Eq. (5), the first coefficient satisfies $A(t) = c_1(t) = \mathbf{w}^\dagger_1 \mathbf{R}(t)$ (we have set $c_1(0)$ to 1), and the second coefficient satisfies $A(t) b(t) = c_2(t) = \mathbf{w}^\dagger_2 \mathbf{R}(t)$. It follows that $b(t)$ is the ratio

$$b(t) = \frac{c_2(t)}{c_1(t)} = \frac{\mathbf{w}^\dagger_2 \mathbf{R}(t)}{\mathbf{w}^\dagger_1 \mathbf{R}(t)}. \tag{20}$$

The coordinate $b(t)$ and the activity basket $\mathbf{R}(t)$ vary by country. We now account for this in our notation by letting b_c be the coordinate associated with country c, and expanding the vector $\mathbf{R}$ into the $n_c \times n_p$ matrix of abilities $R = [R_{cp}]$ across countries and products. In index form, $b_c(t)$ is

$$\begin{aligned}
b_c(t) &= \frac{[R(t)\mathbf{w}_2]_c}{[R(t)\mathbf{w}_1]_c} \\
&= \frac{\sum_p R_{cp}(t) w_{p2}}{\sum_p R_{cp}(t) w_{p1}}.
\end{aligned} \tag{21}$$

Next we use three results to write the b coordinate in a different form. First, in the denominator, we use the fact that $w_{p1} = \pi_p$. Second, in the numerator, we exploit a relationship between the left and right eigenvectors of L_Φ (see Supplementary Note 13),

$$w_{p2} = \pi_p v_{p2}. \tag{22}$$

Applying these changes yields

$$b_c(t) = \frac{\sum_p R_{cp}(t)\pi_p v_{p2}}{\sum_p R_{cp}(t)\pi_p}. \tag{23}$$

Finally, we insert the definition of PCI*, $\mathrm{PCI}^*_p \equiv v_{p2}$. The resulting expression has a structure that closely resembles that of Eq. (8); defining $\mathrm{ECI}^*_c(t) \equiv b_c(t)$, we have

$$\mathrm{ECI}^*_c(t) = \sum_p \left(\frac{R_{cp}(t)\pi_p}{\sum_{p'} R_{cp'}(t)\pi_{p'}} \right) \mathrm{PCI}^*_p. \tag{24}$$

Like the conventional ECI (Eq. (8)), the ECI* is an average of PCI*s. The conventional ECI uses uniform weights for all of the nonzero elements of the binarized matrix M. The ECI*, in contrast, uses non-uniform weights based on the elements of the unbinarized matrix R, and adjusted by the ergodic probabilities $\boldsymbol{\pi}$. (See Supplementary Note 14 for an interpretation and potential benefits of these weights).

Description of data

We use UN Comtrade data[47] available at Harvard Dataverse[62]. Trade data are reported twice: once as exports by the exporting country and once as imports by the importing country. The data cleaning corrects trade flows to increase the consistency between importer and exporter records of the same flow, and further corrects reported values using an index of reliability based on the consistency of reported values over time. The data provide total export volumes of products for 249 countries over the period 1962–2018 (11,544 export baskets observed across all regions and years). We analyze these data at the 3-digit SITC product level (235 product categories).

Calculation of relatedness network and eigenvectors

To obtain our results for country evolution in Fig. 3 we first compute a matrix of relatedness between products using Eq. (18). We fix Φ in the initial year of our data (1962) in accordance with the idea, inherent in the PoR, that economies diversify across a fixed (or at least slow-moving) space of related activities. From Φ we compute the row-normalized version of this matrix $\tilde{\Phi}_{pp'} = \Phi_{pp'}/\sum_{p''}\Phi_{pp''}$, whose right ($\mathbf{v}_\mu$) and left ($\mathbf{w}_\mu$) eigenvectors are the basis for the remainder of the analysis. Let Π be the diagonal matrix formed from the elements of $\boldsymbol{\pi}$. We normalize each left eigenvector such that $\| \mathbf{w} \| \equiv \sqrt{\mathbf{w}^T \Pi^{-1} \mathbf{w}} = 1$ and each right eigenvector such that $\| \mathbf{v} \| \equiv \sqrt{\mathbf{v}^T \Pi \mathbf{v}} = 1$. (See Supplementary Note 13 for additional discussion.) This choice of norms is just a convention and is not consequential, as different choices will simply rescale the coordinates projected onto the eigenvectors.

Use of revealed comparative advantage

Following the literature, our estimation of a country's ability in a product is based on the Balassa index of Revealed Comparative Advantage (RCA)[38],

$$R_{cp} \equiv \frac{\theta_{cp}}{\theta_p} \tag{25}$$

where θ_{cp} is the share of country c's exports composed of product p, and θ_p is the share of p in world exports. The field of economic complexity has been marked by the use of RCAs in its methods as a means to filter out noise and reveal significant exports. RCAs are described in various ways as measures of inferred economic comparative advantage[38], technological advantage[63], competitiveness[51], and indicators of being an effective exporter[15]. When particular economic conditions can be assumed[39,40], it has been shown that RCAs correctly rank the relative domestic prices of products across countries in the absence of trade. In practical terms, the RCA adjusts raw export volumes by normalizing away two overall size effects: variation from differences in the size of economies (more precisely, the size of total exports) and variation because some products are exported in larger volumes than others. The RCA represents a simple approach to removing these size effects, whereby export activity is normalized by both a country's total exports across products and the product's share in world exports. Besides exports, RCAs have also been used to characterize ability in other economic activities, where the meanings of θ_{cp} and θ_p are adjusted appropriately. The quantity Eq. (25) also developed independently in economic geography, where it is known as the Location Quotient, and has been motivated on information-theoretic grounds as an instance of pointwise-mutual information[64].

While RCAs are a widely used standard, they do have potential side effects. As one example, an increase in a location's output of one product may decrease its RCA in other products. While this effect tends to be small, it can be exacerbated when RCAs are binarized around a threshold (typically $R_{cp} = 1$). Binarization is frequently done to generate a simplified picture of what products an economy exports competitively, and to curtail outlier effects from RCAs, which tend to be heavily skewed. In this case, a product beginning with an RCA just above the threshold may realize a new RCA below the threshold, with the counter-intuitive result that the economy appears to 'lose' its ability to effectively export a product (even though its exports of the product have not changed). (This effect does not impact our findings. We study a continuous-time model where elements of $R(t)$ are not binarized and evolve continuously. Binarized RCAs could be used in computing the proximity matrix Φ, where we tried three methods used in the literature (Supplementary Note 6). Two of these, Φ^P and Φ^M, use thresholded RCAs while the third, Φ^C, does not. We do not find large differences in our results, whichever matrix is used. For an extensive empirical investigation of different specifications of the proximity matrix, see Li & Neffke 2024[65]).

Other quantities have also been used to attempt to avoid some of the drawbacks of RCAs. For example, Revealed per Capita Advantage (RpCA) replaces the total export output of a country with population[66], which removes the impact of changes in output of one good on others. As another approach, Brummitt et al.[35] use a regression model to construct a country's expected export volumes in each good. These alternatives represent other ways to put export volumes on a common scale. Here, we deliberately use the conventional RCA, since this is what is most frequently used in analyses of the PoR, but the ideal approach remains debated[63].

Calculation of ability vectors R and (A, b) coordinates

For each country c, product p, and year, we computed the RCA using Eq. (25) with $\theta_{cp} = X_{cp}/\sum_{p'} X_{cp'}$ and $\theta_p = \sum_{c'} X_{c'p}/\sum_{c',p'} X_{c'p'}$, where X_{cp} is the value of c's exports in product p. RCAs are heavy-tailed and are commonly transformed to weaken the effect of extreme values (see, for instance,[16,35,45,46]). Here, we transform them using the function $R = g(R') = \alpha \log(1 + R'/R_0)$. The constant $\alpha = [\log(1 + 1/R_0)]^{-1}$ is included for convenience so that an RCA of 1 is mapped to 1 on the transformed scale. This function behaves linearly for small values of R' and logarithmically for large values, and thus achieves the goal of weakening the effect of very large RCAs while also handling RCAs that are identically zero (which occurs in the many instances where a country has no exports of a product). The parameter R_0 (which sets the transition between the linear and logarithmic regimes) is tuned to maximize the amount of variance in export baskets that the b coordinate explains across time and countries, as we discuss in the next section. Our results can be reproduced using our obtained value $R_0 = 0.115$. We then calculate for each

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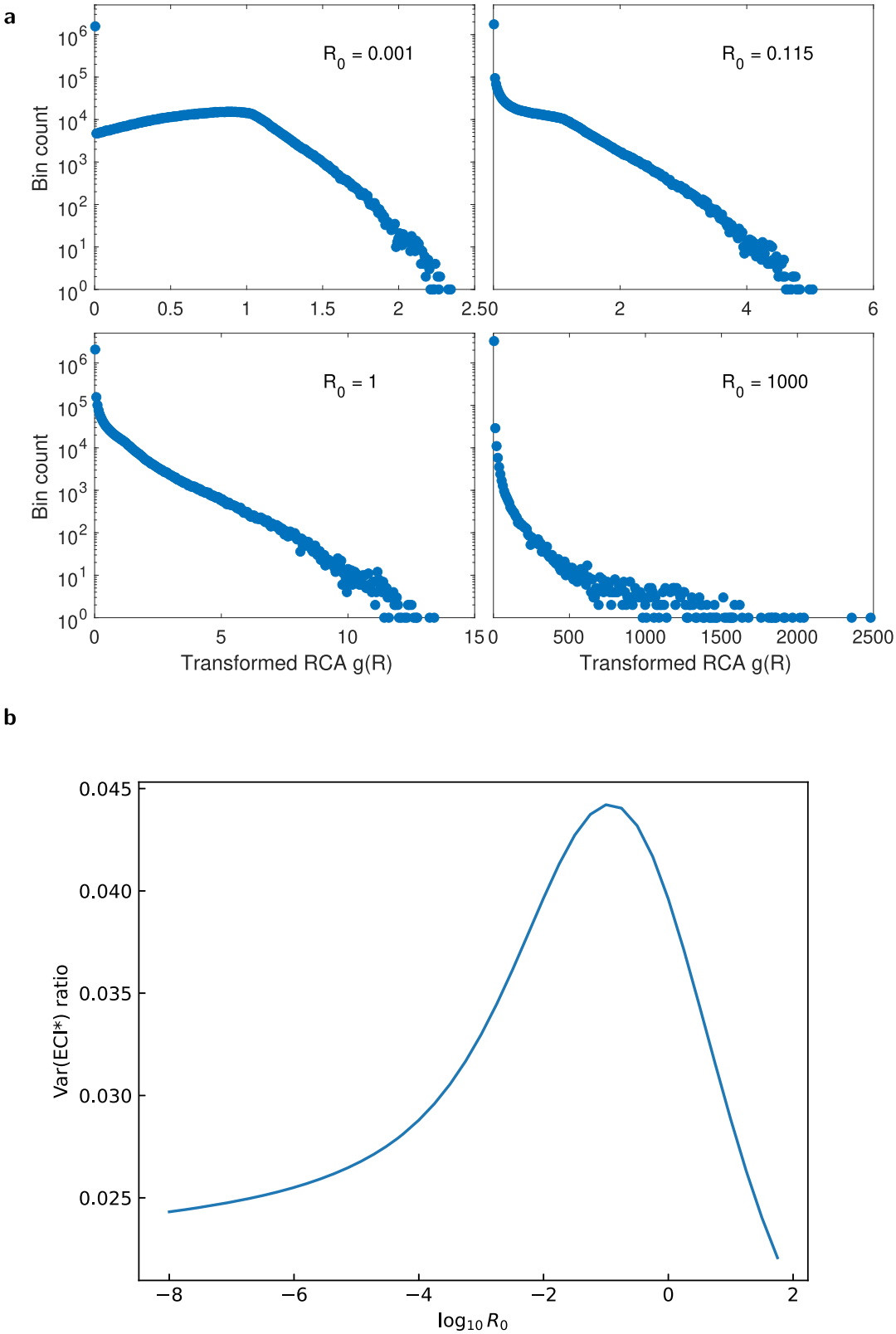

Fig. 6 | Transformation of revealed comparative advantage (RCA). The RCA expresses the overrepresentation of a product in a country's export basket (relative to the product's importance in world trade). It is defined as the share of the product in the country's total export volume divided by the product's share in global exports. The distribution of RCAs is highly skewed, with underrepresentation mapping into the interval [0,1) and overrepresentation onto the interval (0, ∞). Larger RCAs signal that a given country has an apparent relative competitive strength in exporting a given product. We use a monotonic transformation to reduce the distributional skew in these country-product level abilities. **a** Distribution of transformed RCAs $g(R')$ for different values of R_0. Small values of R_0 weaken the expression of the largest RCAs; large values of R_0 do the opposite. (To make the transformation easier to interpret, we divide the transformation function $g(R')$ by the constant $\log(1+1/R_0)$ so that a raw RCA $R'=1$ consistently maps to a transformed RCA $R=1$ for all values of R_0.) **b** Ratio of variance described by variation in the ECI* (Eq. (31)) to total data variability $\sum_a \sigma_a^2(R_0)$. We choose R_0 such that this ratio is maximized. That is, we rescale RCA in a way that maximizes the explainability of global trade patterns.

country c and time t the RCA vector $\mathbf{R}_c(t)$ on this transformed scale and use these vectors to compute the A and b coordinates. The A coordinate for country c and time t is computed as $A_c(t) = \boldsymbol{\pi}^T\mathbf{R}_c(t)$, where $\boldsymbol{\pi} = \mathbf{w}_1$ is the first left eigenvector. The b coordinate is computed as $b_c(t) = \mathbf{w}_2^T\mathbf{R}_c(t)/\mathbf{w}_1^T\mathbf{R}_c(t)$, which is equivalent to Eq. (9).

Transformation of revealed comparative advantage

As noted above, we transform raw RCAs with the function $g(R') = \log(1+R'/R_0)/\log(1+1/R_0)$. The parameter R_0 sets the transition between the linear and log regimes of the transformation, and modulates the expression of extreme values. High values of R_0 allow large RCAs more expression, while low values of R_0 suppress them (Fig. 6a). A given value of R_0 leads to a given activity vector $\mathbf{R}(t;\ R_0)$ and a normalized activity vector $\mathbf{r}(t;\ R_0) \equiv \mathbf{R}(t;\ R_0)/A(t)$, which has the following representation in the eigenvector basis:

$$\mathbf{r}(t;R_0) = \sum_{\mu} b_\mu(t;R_0)\mathbf{v}_\mu = \mathbf{1} + \sum_{\mu\geq 2} b_\mu(t;R_0)\mathbf{v}_\mu. \tag{26}$$

We choose R_0 to maximize the variance explained across countries and time by the directional vector $\mathbf{v}_2 = \mathbf{PCI}^*$, or equivalently, the variance explained by the coordinate $b_2(t;\ R_0) = b(t) = \mathrm{ECI}^*(t)$. Our procedure is related to principal component analysis. The difference is that a PCA uncovers a set of variance-maximizing directions in data, while here, the directions are given to us beforehand (the eigenvectors of the Laplacian L_Φ). Otherwise, we still ask how much variance is explained by data along the particular direction $\mathbf{v}_2 = \mathbf{PCI}^*$ for different values of R_0 and, like PCA, select a value that maximizes data variability described by this direction.

Let $\mathbf{r}_i$ be an observation of a normalized activity basket, and let $X = [\mathbf{r}_i]$ be the $n_p \times N$ data matrix of these observations across countries and years $i \in 1\ldots N$. The eigendecomposition Eq. (26) corresponds to the matrix factorization

$$X(R_0) = VB(R_0), \tag{27}$$

where V is the $n_p \times n_p$ matrix whose columns are L_Φ's right eigenvectors $\mathbf{v}_\mu$ and B is the $n_p \times N$ matrix whose ith column gives the coordinates of the ith observation in the eigenvector basis. The variances and covariances of the data in the directions of each vector in V can be computed as follows. Let $X_c = X\Theta$ be the centered (that is, de-meaned) data, where $\Theta \equiv I_N - \frac{1}{N}\mathbf{1}_N\mathbf{1}_N^T$ is a centering matrix. The data covariance matrix is

$$\begin{aligned} C(R_0) &= \frac{1}{N}X_c(R_0)X_c(R_0)^T \\ &= \frac{1}{N}X(R_0)\Theta\Theta^T X(R_0)^T \\ &= \frac{1}{N}VB(R_0)\Theta\Theta^T B(R_0)^T V^T \\ &= \frac{1}{N}VB_c(R_0)B_c(R_0)^T V^T \\ &= V\tilde{C}(R_0)V^T, \end{aligned} \tag{28}$$

where $B_c = B\Theta$ is the matrix of centered data coordinates in the V basis. The matrix $\tilde{C} = \frac{1}{N}B_cB_c^T$ is the covariance matrix of the data in the coordinate system given by the eigenvectors of the dynamical model. (As further comparison with PCA, recall that in PCA, the data is expressed in a basis V that diagonalizes the covariance matrix. In such a basis $\tilde{C}$ would be diagonal. Here, we are expressing the data in a pre-determined basis given by the eigenvectors of L_Φ, and $\tilde{C}$ will have non-zero off-diagonal elements (non-zero covariances in the new coordinates)).

Let W be the $n_p \times n_p$ matrix whose columns are L_Φ's left eigenvectors $\mathbf{w}_\mu$. Exploiting the biorthogonality of left and right eigenvectors $W^\dagger V = I$, Eq. (28) can be solved for $\tilde{C}$ (the covariances in the V basis) as

$$\tilde{C}(R_0) = W^\dagger C(R_0)W. \tag{29}$$

In particular, the variance of activity baskets in the direction $\mathbf{v}_2 = \mathbf{PCI}^*$ is

$$\begin{aligned} \mathrm{Var}(\mathrm{ECI}^*;R_0) &= \tilde{C}_{22}(R_0) \\ &= \mathbf{w}_2^T C(R_0)\mathbf{w}_2. \end{aligned} \tag{30}$$

The left eigenvectors were normalized with the weighted 2-norm $\|\mathbf{w}\| \equiv \sqrt{\mathbf{w}^T\Pi^{-1}\mathbf{w}}$, while principal components are typically normalized with a standard 2-norm. To remove the scale effect this creates, we factor out the standard 2-norm from the left eigenvectors, computing the variance with $\hat{\mathbf{w}}_2 \equiv \mathbf{w}_2/\|\mathbf{w}_2\|_2$:

$$\widehat{\mathrm{Var}}(\mathrm{ECI}^*;R_0) = \hat{\mathbf{w}}_2^T C(R_0)\hat{\mathbf{w}}_2. \tag{31}$$

For intuition, we could write $\widehat{\mathrm{Var}}(\mathrm{ECI}^*;R_0)$ in terms of the variances explained by principal components. Let Z be the $n_p \times n_p$ matrix whose columns are principal components. Inserting $I = ZZ^T$ above leads to

$$\widehat{\mathrm{Var}}(\mathrm{ECI}^*;R_0) = \sum_a \left(\hat{\mathbf{w}}_2^T\mathbf{z}_a(R_0)\right)^2\sigma_a^2(R_0). \tag{32}$$

Thus, the data variance in the direction $\mathbf{v}_2 = \mathbf{PCI}^*$ can be written as a weighted sum of the variances of the principal components, with each principal component weighted by its projection $(\hat{\mathbf{w}}_2^T\mathbf{z}_a(R_0))$ onto $\mathbf{v}_2$ in the non-orthogonal basis V. We score the variance that the coordinate $b = \mathrm{ECI}^*$ explains using the ratio of Eq. (31) to the total data variability $\sum_a\sigma_a^2(R_0)$. We calculate this ratio for various values of R_0, finding a peak near $R_0 = 0.115$ (Fig. 6b).

Data availability

The trade data used in this study are freely and publicly available in the Harvard Dataverse database under accession code https://doi.org/10.7910/DVN/H8SFD2, version 4.0[67] [https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/H8SFD2&version=4.0].

Code availability

The code used for the study have been deposited at https://github.com/complexly/por-structuralchange(https://doi.org/10.5281/zenodo.7761169)[68].

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Author contributions

J.M., Y.L., A.G-L., and F.N. conceived and designed the study. J.M., Y.L., and F.N. prepared the data and developed code. J.M., Y.L., A.G-L., and F.N. analyzed the data. J.M. and Y.L. produced the figures. J.M. and F.N. wrote the manuscript. J.M., Y.L., A.G-L., and F.N. edited and reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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